SMoPE: Research Notes & Future Work

Consolidated record of the presentation limitations, Task 1 and Task 2 results, proposed improvements, and next research steps

Purpose. This document combines the two sets of notes shared in the conversation: (1) the limitations and research-gap summary, and (2) the detailed review of the 16-slide SMoPE presentation and proposed experiments. It is intended as a reference for restarting the project later.

Important distinction. The reported numbers below are the experimental results supplied in the notes. Proposed methods and expected benefits are research ideas or hypotheses; they are not established results until tested.

1. Project overview and central research problem

The project studies improvements to Sparse Mixture of Prompt Experts (SMoPE) for continual learning. The two current experimental directions are:

• **Task 1 — Expert-Choice Prompt Routing:** improve expert utilization by changing how inputs are assigned to prompt experts.

• **Task 2 — Orthogonal Prompt Expert Specialization:** reduce similarity between prompt-expert prefix keys through orthogonality regularization.

The central issue is that better-looking routing balance or parameter-space diversity does not automatically produce better classification accuracy or continual-learning performance.

2. Main limitations to keep in mind

Limitation	Why it matters	Possible direction
Expert imbalance	Input-wise Top-K routing can repeatedly select popular experts and underuse others.	Balanced or coverage-aware routing; monitor coverage as well as balance.
Expert redundancy	Different prompt experts may learn similar prefix-key directions.	Orthogonal prompt regularization, guided by actual routing behaviour.
Fixed Top-K routing	Every input receives the same number of experts regardless of difficulty.	Adaptive K or dynamic expert allocation.
No explicit functional diversity	Different prompt keys do not necessarily produce different attention outputs.	Function-level or routing-aware diversity loss.
Limited task-aware protection	Prototype loss can help preserve old knowledge but does not guarantee stable earlier-task performance.	Prototype-preserving routing or targeted stability regularization.
Frozen backbone	Prompt adaptation does not allow the underlying ViT features to learn new representations.	Selective adapters or low-rank backbone adaptation.
Routing sensitivity	Expert selection may change when input characteristics or score distributions shift across tasks.	Routing-consistency regularization and stability analysis.
Limited evaluation scope	The presented experiments mainly use CIFAR-100; generalization to other continual-learning settings is not established.	Evaluate on additional benchmarks such as ImageNet-R or CUB-200.

The first two limitations are directly connected to the current experiment visualizations. The remaining items are architectural concerns or research hypotheses that need targeted experiments. SMoPE already includes mechanisms such as adaptive noise, router loss and prototype loss; therefore, do not describe the original method as having no

balancing, specialization or forgetting-related mechanisms.

3. Experimental evidence — current Task 1 and Task 2

3.1 Task 1 — Expert-Choice Prompt Routing

Metric	Original SMoPE	Expert-Choice
FAA (%)	89.71	87.42
CAA (%)	92.41	90.44
Active experts/input	5.00	4.70
Inputs with zero experts	0%	66.53%
Expert utilization CV	1.998	0.000
Prompt MACs/image	4.68M	4.40M

Source: presentation slide 11, as reported in the project review.

Interpretation: Expert-choice routing achieves perfectly balanced expert workloads (CV = 0), but FAA falls by 2.29 percentage points and CAA by 1.97 points. Most importantly, 66.53% of inputs receive zero experts. This is a plausible contributor to the accuracy decline, but a controlled experiment is needed to establish the causal contribution.

Key caution: zero workload CV is achieved by construction under equal expert capacities. It is not sufficient evidence of better routing quality.

3.2 Task 2 — Orthogonal Prompt Experts

Orthogonality strength μ	FAA (%)	CAA (%)	Mean absolute cosine
Original	89.71	92.41	0.594
0.001	89.90	92.54	0.139
0.1	89.65	92.40	0.072
1.0	89.74	92.44	0.053

Source: presentation slide 13, as reported in the project review.

At μ = 0.001, FAA improves by 0.19 points and CAA by 0.13 points. At μ = 1.0, cosine similarity is much lower, but accuracy does not improve meaningfully. This indicates that geometric diversity between prefix keys does not automatically create useful expert specialization.

The earlier summary also describes high pairwise cosine similarity in the original prefix keys. The numerical table above is the detailed result set to use when discussing the measured trend.

4. Detailed future work — Task 1

4.1 Coverage-Aware Expert-Choice Routing (highest priority)

Modify expert-choice routing so every input receives at least one expert, then allow experts to compete for additional inputs subject to capacity limits. This directly targets the 66.53% zero-expert rate.

Current assignment (as described in the notes):

```
A[b,e] = 1 if input b is among the Top-C scores for expert e; otherwise 0.
```

Proposed coverage-constrained assignment:

```
Maximize Σ_b Σ_e A[b,e] S[b,e]

Subject to: Σ_e A[b,e] ≥ 1 for every input b; and Σ_b A[b,e] ≤ C_e for every expert e.
```

Each input is guaranteed coverage while expert capacities limit overload. Feasibility requires total available expert capacity to be at least the batch size; otherwise capacity must increase or a fallback route is needed.

Experiment: compare original expert-choice, coverage-aware expert-choice, and capacity factors 0.5, 1.0, 1.25 and 1.5. Report zero-expert percentage, FAA, CAA, utilization CV and compute.

4.2 Adaptive capacity instead of fixed capacity

Experts may have different routing demands. Use learned or utilization-dependent capacity rather than assigning every expert the same number of inputs.

```
C_e = clip(round(C_0 × w_e), C_min, C_max), where w_e is a normalized importance or demand
score for expert e.
```

Keep a total capacity budget so assignments remain feasible. Compare fixed and adaptive capacity using FAA, CAA, utilization CV, zero-expert percentage and computational cost.

4.3 Hybrid input-choice and expert-choice routing

• Assign one expert to each input using input-wise Top-1 routing.

• Use remaining capacity for expert-choice assignments.

• Preserve the original score-based prompt weighting.

• Enforce a maximum number of experts per input.

This may combine input coverage with more even utilization. Pure Top-1 can overload popular experts, so use capacity-constrained matching when strict capacity limits matter. Compare Top-1 fallback, constrained matching and fully hybrid routing.

4.4 Task-aware expert routing

Use existing SMoPE prototype information to encourage routing that preserves experts useful for previously learned tasks. One proposed score is:

```
S'[b,e] = S[b,e] + λ_p × sim(z_b, p_e)
```

Here z_b is the current input representation and p_e summarizes the representation region associated with expert e. Test per-task accuracy, forgetting and expert-assignment stability after each new CIFAR-100 task. This could reduce interference without storing historical training images, but that benefit must be verified.

5. Detailed future work — Task 2

5.1 Routing-aware orthogonality (highest priority)

The current orthogonality loss penalizes pairwise cosine similarity between all prefix-key experts independently of routing behaviour. Instead, apply stronger penalties to experts frequently selected for similar inputs.

```
Co-activation weight: w[i,j] = Σ_b A[b,i]A[b,j] / (sqrt(Σ_b A[b,i] × Σ_b A[b,j]) + ε)
```

```
Routing-aware orthogonality loss: sum over i ≠ j of w[i,j] × (k■_i■ k■_j)², normalized by the sum of the co-activation weights.
```

The goal is to encourage diversity especially among experts serving the same inputs. Experts that never handle similar inputs may already be functionally specialized even if their keys are geometrically similar. This is a testable hypothesis, not a conclusion established by the slides.

5.2 Task-adaptive orthogonality

Vary the regularization strength across continual-learning stages. One example is μ_t = μ_min + (μ_max – μ_min) × (t – 1)/(T – 1). Test increasing and decreasing schedules as well as a constant μ: stronger late regularization could also damage previously learned representations.

Experiment: compare FAA, CAA and forgetting under constant, increasing and decreasing μ schedules.

5.3 Function-level expert diversity

The current loss measures diversity in prefix-key parameter space, not necessarily in actual attention outputs. Add an auxiliary loss that penalizes similar outputs for co-active expert pairs, for example the average squared cosine similarity between f_i(x) and f_j(x) over selected pairs.

This can be more computationally expensive. Apply it selectively and check that useful shared features are not being penalized. Research question: does functional diversity correlate more strongly with continual-learning accuracy than prefix-key orthogonality?

5.4 Prototype-preserving orthogonality

Combine orthogonality with a constraint preserving useful old-task expert representations:

```
L = L_SMoPE + μ L_orth + η L_preserve
```

The preservation loss can compare current expert representations on stored old-task prototypes with their previously saved representations. This is relevant because SMoPE already uses prefix keys as prototypes to help avoid catastrophic forgetting.

6. Other architectural limitations and possible directions

Potential limitation	Possible direction / evaluation
Fixed expert budget over long task sequences	Study whether a finite shared expert pool becomes insufficient as tasks accumulate; evaluate dynamic expert expansion.
Frozen ViT backbone	Compare prompt-only learning with selective adapters or low-rank backbone adaptation.
Routing sensitivity under distribution shift	Measure assignment consistency when input characteristics or score distributions shift across tasks.
No guarantee that key diversity means functional diversity	Compare key-space similarity, co-activation and attention-output similarity.
Old-task performance may still drift	Report per-task accuracy and forgetting, and test prototype-preserving constraints.
Limited benchmark coverage	Extend beyond CIFAR-100 to at least one additional continual-learning benchmark, such as ImageNet-R or CUB-200.

These are candidate limitations and research directions, not all experimentally demonstrated by the current slides.

7. Prioritized implementation plan

Priority	Task	Experiment	Main target
1	Task 1	Coverage-aware expert assignment	Eliminate zero-expert inputs
2	Task 1	Hybrid Top-1 + expert-choice routing	Recover FAA / CAA while retaining balance
3	Task 2	Routing-aware orthogonality	Improve useful specialization
4	Task 2	Task-adaptive regularization	Study and reduce forgetting
5	Combined	Coverage-aware routing + routing-aware orthogonality	Test complementary benefits

Working name for the combined proposal: Coverage-Aware, Diversity-Regularized SMoPE. This is a descriptive working name for the proposed extension, not an established novel method.

8. Experiments and reporting checklist

• **Task 1 capacity-factor ablation:** plot FAA, CAA, expert utilization and zero-expert coverage against capacity factor.

• **Task 2 orthogonality analysis:** plot cosine similarity against FAA and CAA; show that stronger orthogonality does not consistently improve accuracy.

• **Loss ablations:** separate the contribution of each new loss from the existing SMoPE objectives.

• **Multiple seeds:** run at least 3–5 seeds and report mean and standard deviation.

• **Continual-learning evaluation:** provide a full per-task accuracy matrix and forgetting metrics, not only FAA and CAA.

• **Fair comparison:** keep backbone, task order, compute budget and evaluation protocol identical across methods.

• **Routing diagnostics:** report zero-expert percentage, active experts per input, utilization CV, assignment stability and routing heatmaps.

• **Efficiency:** report prompt MACs or comparable computational overhead.

9. Presentation improvements to revisit

• **Slide 10:** clarify that expert-choice routing depends on other samples in the minibatch, not literally on future tasks or future tokens. Add batch-size sensitivity.

• **Slide 11:** highlight the 66.53% zero-expert rate and accuracy decline; this is the central negative result motivating the next method.

• **Slide 12:** add a heatmap comparing original routing, expert-choice and proposed coverage-aware routing.

• **Slide 13:** explain why $\mu = 0.001$ is currently the best observed setting for accuracy, even though larger μ gives lower cosine similarity.

• **Slide 15:** add a quantitative plot of cosine similarity against FAA/CAA and, if possible, per-task forgetting.

10. Recommended research direction

Coverage-Aware Routing + Routing-Aware Orthogonality

The combined research question is: **How can sparse prompt experts be assigned efficiently while ensuring every input is covered and co-active experts learn complementary representations?**

This is a more coherent direction than presenting two independent modifications. However, novelty relative to existing coverage-constrained MoE routing, expert-choice routing and diversity-regularized MoE methods still requires a targeted literature review.

Recommended order: first fix Task 1's uncovered-input problem; next test routing-aware orthogonality for Task 2; finally combine both and evaluate whether their benefits are additive on CIFAR-100 and at least one additional continual-learning benchmark.

The immediate research milestone is not simply improving heatmaps. It is demonstrating that better routing coverage and meaningful expert diversity translate into higher FAA/CAA, lower forgetting and reasonable computational overhead.

Reference note: This PDF consolidates the user-provided limitations summary and the attached detailed SMoPE review. It does not independently verify the underlying presentation, source paper, equations, or experimental results. Check the original slides and implementation before using the values or claims in a manuscript.